THE LAWS OF SOFTWARE TEAMS

Brooks' law

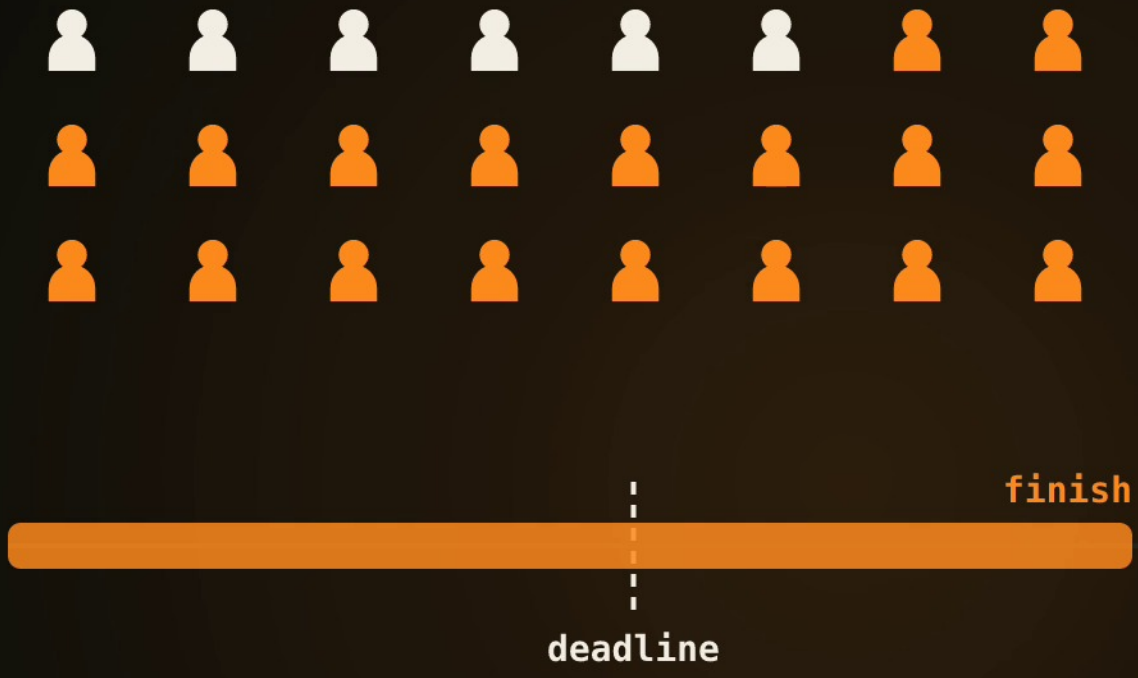
Adding people to a late software project makes it later

swipe →

Ivan Rykov

IBM, 1960s · OS/360

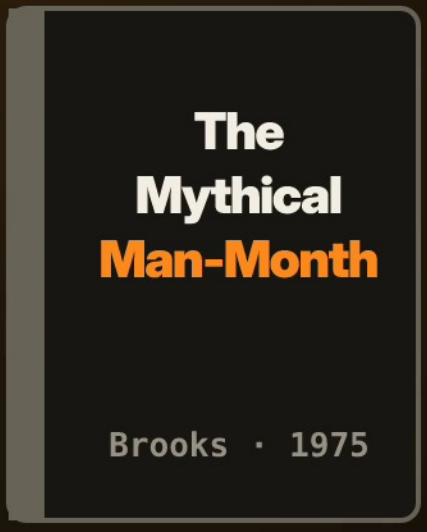
Late? Add people
Even later



peak team: ~1,000 people · Fred Brooks, project manager

“Adding manpower to a late software project makes it later”

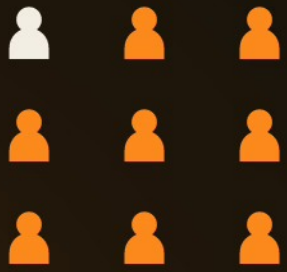
Brooks’ law



Some work splits
Some doesn't



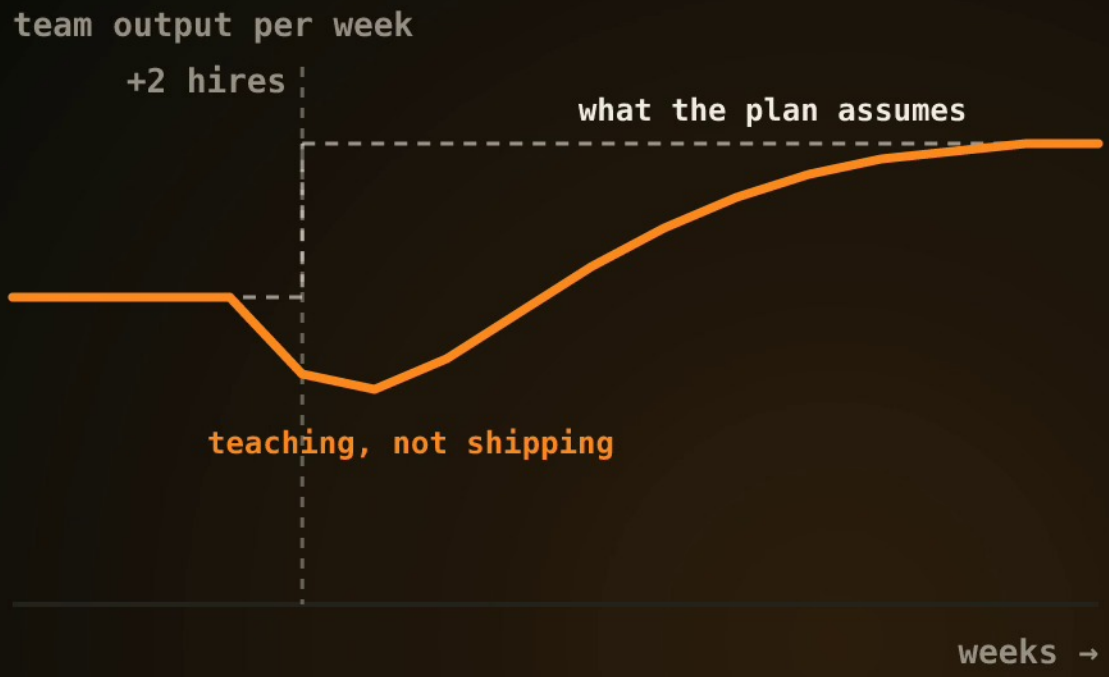
10 rooms, 10 cleaners
`done in 1 hour`



a baby, 9 people
`still 9 months`

Nine women can't make
a baby in one month

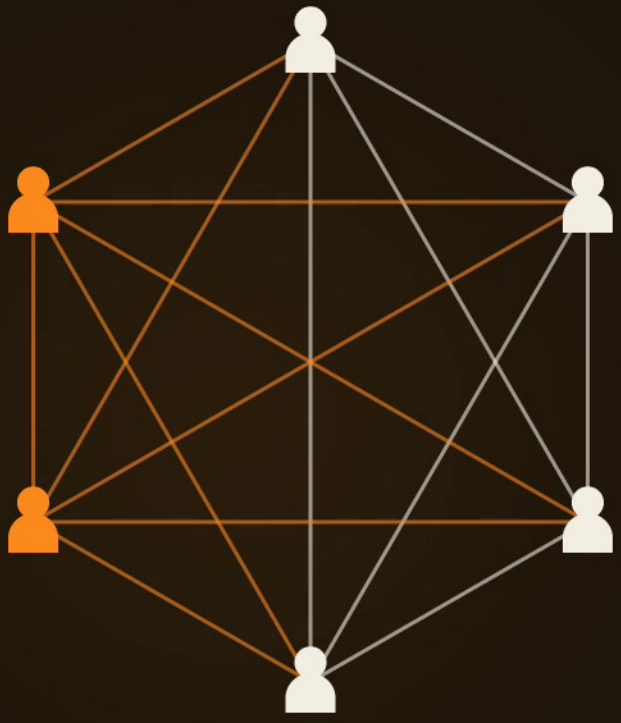
New people slow you down first



Your best engineer becomes a teacher

toy model

Every hire adds conversations



6 people → 15 lines

+50% people → +150% lines · n(n−1)/2

Brooks himself called it an outrageous oversimplification

1. **Add people early**
   not when you're already late

2. **Give clear, separate pieces**
   testing, docs, a module of their own

Already late?

- Cut scope
- Move the date
- Remove blockers
- ~~Throw bodies at it~~

People and months don't trade

```
6 people × 2 months
        ≠
12 people × 1 month
```

The man-month is a myth

WHAT I TAKE FROM THIS

When management pushed me to add people to late projects, it did not help much. Communication and frustration grew, the work did not split well, and in most cases the system was not ready for more people. What works better:

1. **I add people early, when an important project has risks and unknowns. They take delegated tasks and learn the project before problems start**
2. **I take people from inside the company, preferably with context. With good context and experience, they can take any part. People without context get tasks that have clear steps, do not depend on the domain and can be done end to end**

WHAT I TAKE FROM THIS

3. Usually someone more senior introduces them to the project: me, the project lead together with me, or the person who delegates the task
4. If the project is late anyway, there is one rare case when we cut nothing. In large companies, a deadline can come down through several management levels, and each level adds its own buffer. Then we can move the team's date a little later and still meet the real deadline

WHAT I TAKE FROM THIS

5. **Otherwise, we cut. We go through MoSCoW again, drop the "could have" items, remove some edge cases from the scope and focus on the happy path**
6. **The team becomes a working group busy only with this project. Syncs and short planning are fine, but not long daily discussions. Nobody pulls them into other work**

Sources: Fred Brooks, "The Mythical Man-Month", 1975 · Fast Company, 2013

next: the Ringelmann effect